

JET FOR A NEW GENERATION A full-scale model of the AMCA at Aero India 2025, Bengaluru

GE F414 engines in India in the 98 kN thrust class with 80 per cent transfer of technology are still being negotiated. The GE F414 is to power the Tejas Mk 2 fighter, as well as the first batch of the AMCA. The penalties imposed by India on GE are said to have complicated the talks.

Some aeroengine experts maintain that the GE F414 engine will be inadequate for both the AMCA and the Tejas Mk 2. "The AMCA is designed for an engine with 120kN thrust," one scientist tells INDIA TODAY. Similarly, the GE F414 may be inadequate for Tejas Mk 2—a medium-weight fighter of 15-18 tonne range carrying a five-tonne combat payload.

There is another issue that goes against a new deal involving the GE F414. Even though 80 per cent tech transfer is assured for the GE F414s meant for the LCA Mk 2 and early AMCAs, sceptical Indian designers are aware that the Americans are loath to share high-end technology despite selling almost $20 billion (Rs 1.77 lakh crore) worth of defence material to India in the past 15 years. But France, which supplies a wide range of military platforms to India, is keen to enhance its strategic partnership.

The engine Safran will develop with India will be a more powerful version of the M-88, the 75 kN engine that powers the Rafale. And Safran will help DRDO/ HAL with full transfer of knowhow, from creating assembly lines, testing facilities and helping with complex metallurgy. Crucially, all parts are going to be sourced from within India. The reason why the estimated cost of this project is much higher than both the GE F414 and the GE F404 deals is that the plan with Safran is to develop a much more potent engine from the ground up—including building manufacturing and testing facilities—as well as the promise of full transfer of technology and intellectual property rights.

However, 'technology transfer' often doesn't materialise to its promised degree. For example, HAL and Safran have jointly produced the Shakti engine for Indian helicopters since 2003 under a similar contract, but less than 50 per cent of parts are made in India with full tech transfer. The Sukhoi Su-30MKI programme with Russia, which promised 60 per cent indigenous content, ended up with only about 13 per cent Indian-made parts. "One should know what to ask for...unless you ask for it, the original equipment manufacturer will never share something that took decades to develop," says a scientist.

THOUGH THE COLLABORATION WITH SAFRAN FOR AN ENGINE FOR AMCA IS A NECESSITY, EXPERTS SAY INDIA CANNOT EXPECT THE FIRM TO HAND OVER A COMPLETE DESIGN, IP RIGHTS

A LONG HAUL

The struggle to produce a jet engine poses something of a paradox. How is it that India can make world-class nuclear submarines and cruise/ ballistic missiles, but not jet engines? Defence analyst Girish Linganna points out that jet engine development takes decades of research, testing and analysis of failure data. Modern jet engines face rapid acceleration and deceleration, have to withstand high temperature and travel several times the speed of sound.

In 1986, India launched the Kaveri engine project, aiming to develop a homegrown jet engine to power its LCA Tejas. T. Mohan Rao, former director of the GTRE, says that India's indigenous engine fell short by only about 10 per cent in thrust. "A gas-turbine engine has nearly 30,000 components, each performing under extreme temperatures. Even nations like Russia, the US or France took 30 to 35 years to master this technology," he says, adding that critical technologies such as single-crystal turbine blades and specialised bearings were not available in India at the time. "But no country was willing to share these core technologies."

India developed nine prototypes of the Kaveri, but none met the parameters to power a fighter jet, despite an expense of over Rs 3,000 crore. The project was shelved in 2011.

The impending collaboration with Safran, thus, is a necessity for India's aerospace industry. India can't remain dependent on foreign suppliers. Experts also say India should not expect Safran to hand over a complete design/ IP rights; Indian development must begin with its own design initiative. A consortium approach with private sector agility, public sector infrastructure and foreign knowledge is essential. However, real success lies in how much technology stays with India when French engineers fly back home. ■